Mohammed Sazzad Yousuf Sourab

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Professional Summary

Graduate researcher and engineering educator specializing in AI, Explainable AI, Quantum Machine Learning, and Energy Efficient Computing. Published author with peer-reviewed work in medical image analysis and deep learning. Experienced university lecturer with a background in telecommunications engineering, outcome-based education, and student mentorship.

Education

Lamar University — Beaumont, Texas
Master of Science (M.S.) in Electrical and Computer Engineering. — *Expected May 2027*
Relevant Coursework: AI Accelerators, Energy Efficient Computing, Computer Vision, Quantum Machine Learning

Chittagong University of Engineering & Technology (CUET) — Chittagong, Bangladesh
Bachelor of Science (B.Sc.) in Electronics and Telecommunication Engineering. GPA: 3.71/4.00 (3.94/4.00 in final 60 credits). — *April 2022*

Experience

Graduate Research Assistant — January 2026 – Present
Lamar University, Department of Electrical and Computer Engineering — *Beaumont, Texas*
- Conduct graduate research in AI-driven systems, energy efficient computing, and quantum machine learning (QML)
- Benchmark hybrid quantum models (QCNN, QuanvNet, QViT) on TACC Lonestar6 H100/A100 GPUs, measuring accuracy alongside throughput, GPU power, and training energy
- Develop interpretable machine learning models with focus on explainable AI and power profiling in AI accelerators
- Collaborate on research projects in medical image analysis, computer vision, and intelligent engineering systems

Lecturer — August 2022 – December 2025
Daffodil International University, Department of Information and Communication Engineering — *Dhaka, Bangladesh*
- Delivered undergraduate courses in telecommunications, information and communication engineering, and digital systems
- Implemented outcome-based education (OBE) methodologies to enhance student learning outcomes
- Mentored undergraduate students in academic development, research projects, and capstone projects
- Supervised student projects in machine learning, AI applications, and telecommunications

IELTS Instructor — July 2024 – 2025
10 Minute School — *Bangladesh*
- Instruct students in IELTS exam preparation with focus on academic and professional English skills
- Develop instructional materials and assessment strategies for diverse learning needs

Digital Literacy Instructor, Youth Project — January 2023 – December 2023
Consumers Association of Bangladesh (CAB) — *Dhaka, Bangladesh*
- Taught digital literacy skills to youth participants, enhancing their technical competencies
- Developed curriculum focused on practical technology skills for career development

Solution Architect (Cloud) — June 2022 – August 2022
Huawei Technologies Ltd. — *Bangladesh*
- Designed and implemented cloud-based telecommunications solutions for enterprise clients
- Provided technical consultation on cloud migration strategies, scalable network architecture, and solution optimization

Web Developer Intern — December 2020 – June 2021
Mechanic Koi — *Dhaka, Bangladesh*
- Developed and maintained web applications using modern frameworks and technologies
- Gained experience in full-stack development, database management, and project lifecycle

Publications

- Sourab, M. S. Y., Hossain, M. E., Chowdhury, F. A., & Ferdaus, F. (2026). “Benchmarking Quantum Image Dataset Transform Pre-processing on Quantum Architectures.” 2026 IEEE International Conference on Quantum Computing and Engineering (QCE). doi: 10.1109/QCE68830.2026.10127
- Islam, T., Sourab, M. S. Y., & Haque, A. K. M. F. (2026). “View-Aware Reliability Framework for Pneumonia Diagnosis with Calibration Portability, Selective Risk, and XAI Auditing.” 11th IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE).
- Islam, T., Haque, A. K. M. F., Hasan, M., & Sourab, M. S. Y. (2026). “SAFER-RAG: Explainable and Calibrated Hallucination Detection with Selective Prediction under Distribution Shift.” 11th IEEE Asia-Pacific Conference on Computer Science and Data Engineering (CSDE).

- Sourab, S. Y., & Kabir, M. A. (2022). “A comparison of hybrid deep learning models for pneumonia diagnosis from chest radiograms.” Sensors International, 100167. doi: 10.1016/j.sintl.2022.100167
- Sourab, S. Y., Shuvo, H. R., Hasan, R., & Masruf, T. (2021). “Diagnosis of COVID-19 from Chest X-Ray Images Using Convolutional Neural Networking with K-Fold Cross Validation.” 2021 IEEE International Power and Renewable Energy Conference (IPRECON), pp. 1-5. doi: 10.1109/IPRECON52453.2021.9640744
- Chowdhury, M. T. I., Islam, A., Hussain, S. N., Sourab, S. Y., & Haider, M. A. (2021). “An Approach on Detection of Brain Tumor Using Adaptive Network Fuzzy Inference System (ANFIS).” Intelligent Robotics, Mechatronics and Automation Systems (IRMAS 2021), India.

POSTER PRESENTATIONS

- Sourab, M. S. Y., & Ferdaus, F. (2026). “Power-Trace Fingerprinting of AI Models on Modern GPUs: A Side-Channel Methodology for Vision and Language Architectures.” Texas Advanced Computing Center Student Symposium (TACCSTER 2026), Austin, Texas. doi: 10.26153/tsw/65039
- Sourab, M. S. Y., & Ferdaus, F. (2026). “Lost in Telemetry: Measurement Correctness in Power Side-Channel Recovery of Neural Network Architectures on Datacenter GPUs.” International Conference for High Performance Computing, Networking, Storage and Analysis (SC26). *Upcoming.*

PROFESSIONAL MEMBERSHIPS

- The Institution of Engineers, Bangladesh (IEB) - Membership No: M/50739
- Institute of Electrical and Electronics Engineers (IEEE) - Graduate Student Membership No: 94594942 (valid through December 2027)

LEADERSHIP AND ACTIVITIES

- Treasurer, Graduate Student Association (College of Engineering), Lamar University (2026 - 2027)
- President, CUET Photographic Society (March 2021 - June 2022)
- Photographic Secretary, CUET Debating Society (February 2021 - June 2022)
- Photographic Secretary, Robo Mechatronics Association (RMA) Bangladesh (January 2021 - July 2022)
- Photography Secretary, CUET Career Club (December 2020 - August 2022)
- Campus Ambassador, Team Matrix - Elite Hackers (January 2021 - December 2021)
- Student Ambassador, Bangladesh Model United Nations (BANMUN) in Association with UNYSAB (2016)

SKILLS

Research and Analysis: Machine Learning, Deep Learning, Artificial Intelligence, Explainable AI (XAI), Quantum Machine Learning (QML), Computer Vision, Medical Image Analysis, AI Accelerators, Energy Efficient Computing, Quantum Computing, Research Methodology

Programming Languages: Python, C/C++, R, Java, JavaScript, HTML, CSS, PHP, MySQL, Assembly Language (8086), MPI, Slurm

Simulation and Design Tools: Qiskit, MATLAB and Simulink, Google Colab, Cisco Packet Tracer, Cadence, CST Studio, Proteus, Multisim, Ultiboard, Git

Engineering: Telecommunications Engineering, Electronics Engineering, Power Profiling, Cloud Computing

Teaching and Education: University Lecturing, Outcome-Based Education (OBE), Curriculum Development, Student Mentoring

Software and Tools: MS Office 365, Adobe Photoshop, Adobe XD, Adobe Premiere Pro

CERTIFICATIONS

- Machine Learning and AI Foundations: Predictive Modeling Strategy at Scale
- Neural Networks and Convolutional Neural Networks Essential Training
- Generative AI for Educators Certificate

AWARDS & HONORS

- Gold Award - 1st International Poster Competition on Achieving Sustainable Development Goals (SDGs) in Developing Countries (2021)
- Best Paper Award - Big Data and Machine Learning Workshop (BDML 2020), Department of CSE, CUET
- Best Motion Maker - 13th Inter-Department Debate Competition, CUET
- Champion - Informatics Olympiad, 10th Dhaka Residential Model College National Science Fair (DRMCNSF) (2016)
- Government Merit Scholarship - Ministry of Education, Bangladesh (2011, 2016)